

EXPERIMENT 1

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Section: 08

Title

Implementation of Process Creation and Command Execution in Linux using fork(), exec(), and wait() System Calls.

Aim

To develop a C program that accepts a Linux command from the user, creates a child process using fork(), executes the command, waits for the child process, and displays Parent and Child PIDs.

Theory

Process creation in Linux can be performed using the fork() system call. The fork() system call creates a new child process. The child process can execute a Linux command using an exec() family system call such as execlp(). The parent process can wait for the child process to complete using wait(). Process IDs (PIDs) are obtained using getpid() and the parent PID using getppid().

Algorithm

1. Start the program.
2. Accept a Linux command from the user.
3. Remove the newline character from the input.
4. Create a child process using fork().
5. If fork() fails, display an error message and terminate.
6. In the child process, display the Child PID and Parent PID.
7. Execute the entered Linux command using execlp().
8. In the parent process, display the Parent PID.
9. Wait for the child process to complete using wait().
10. Display the child process completion message.
11. Stop the program.

Code

```
k_anusri@DESKTOP-TM20BAQ:~$ cat > main.c <<'EOF'
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    printf("OSSP Experiment 1\n");
    printf("Shell Architecture\n");
    printf("Parent PID: %d\n", getpid());

    pid_t pid = fork();

    if (pid == 0)
    {
        printf("Child Process PID: %d\n", getpid());
    }
    else if (pid > 0)
    {
        wait(NULL);
        printf("Parent Process: Child completed\n");
    }
    else
    {
        printf("Fork failed\n");
    }

    return 0;
}
EOF
```

Program Execution using lscpu

```

k_anusri@DESKTOP-TM20BAQ:~$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:               39 bits physical, 48 bits virtual
Byte Order:                  Little Endian
CPU(s):                      8
  On-line CPU(s) list:       0-7
Vendor ID:                   GenuineIntel
Model name:                  Intel(R) Core(TM) i5-10300H CPU @ 2.50GHz
CPU family:                  6
Model:                       165
Thread(s) per core:          2
Core(s) per socket:          4
Socket(s):                   1
Stepping:                    2
BogoMIPS:                    4992.01
Flags:                       fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat ps256 clflush mmx fxsr ss
                             e sse2 ss ht syscall nx pdpe1gb rdtscp lm arch_perfmon rap_good nopl xtopology cpuid tsc_kn
                             own_freq pni pclmulqdq vmx ssse3 fma cx16 pdcm pcid sse4_1 sse4_2 x2apic avx bswap aes x
                             save avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs ibpb stibp lbrs enhance
                             d tpr_shadow ept vpid ept_ad fsgsbase bml avx2 smep bmi2 erms invpcid rdseed adx smap clfl
                             ushopt xsaveopt xsaves xgetbv1 xsaves vmmi md_clear flush_lld arch_capabilities

Virtualization features:
  Virtualization:            VT-x
  Hypervisor vendor:         Microsoft
  Virtualization type:       full

Caches (sum of all):
  L1d:                       128 KiB (4 instances)
  L1i:                       128 KiB (4 instances)
  L2:                         1 MiB (4 instances)
  L3:                         8 MiB (1 instance)

NUMA:
  NUMA node(s):              1
  NUMA node0 CPU(s):         0-7

Vulnerabilities:
  Gather data sampling:       Unknown: Dependent on hypervisor status
  Ghostwrite:                 Not affected
  Indirect target selection:  Mitigation; Aligned branch/return thunks
  Itlb multihit:              KVM: Mitigation: Split huge pages
  L1tf:                       Not affected
  MDS:                        Not affected
  Meltdown:                   Not affected
  Mmio stale data:            Vulnerable: Clear CPU buffers attempted, no microcode; SMT Host state unknown
  Old microcode:              Not affected
  Rgb file date sampling:     Not affected
  Retbleed:                   Mitigation; Enhanced IBRS
  Spec rstack overflow:       Mitigation
  Spec store bypass:          Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:                 Mitigation; usercopy/swaps barriers and __user pointer sanitization
  Spectre v2:                 Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSB-eIBRS SW sequence; BHI SW lo
                             op /SW SW loop
  Srbds:                     Mitigation; affected:
  Tsx:                        Not affected
  Tsx async abort:           Not affected
  Vmspace:                   Not affected

```

Program Execution using uname

```

k_anusri@DESKTOP-TM20BAQ:~$ uname -a
Linux

```

Program Execution using lsblk

```
k_anusri@DESKTOP-TM20BAQ:~$ lsblk
NAME MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 186.5G  0 disk
└─sdb 8:16    0 159.4M  1 disk
sdc   8:32    0   20G   0 disk [SWAP]
sdd   8:48    0    1T   0 disk /mnt/wslg/distro
```

Program Execution using ps

```
k_anusri@DESKTOP-TM20BAQ:~$ ps
  PID TTY          TIME CMD
  561 pts/2        00:00:00 bash
  883 pts/2        00:00:00 ps
```

Program Execution using top

```
k_anusri@DESKTOP-TM20BAQ:~$ top
top - 04:58:42 up 10 min, 1 user, load average: 0.01, 0.04, 0.00
Tasks: 24 total, 1 running, 23 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7859.6 total, 511.5 free, 174.9 used, 7349.1 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7343.1 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	20184	15048	11624	S	0.0	0.7	0:01.26	systemd
2	root	20	0	3188	2208	2074	S	0.0	0.3	0:00.01	init-systemd(Ub
3	root	20	0	3188	2044	1648	S	0.0	0.2	0:00.00	init
48	root	19	-1	58408	16776	15480	S	0.0	0.2	0:00.29	systemd-journal
83	systemd+	20	0	22416	14080	11984	S	0.0	0.7	0:00.11	systemd-resolve
91	root	20	0	15544	12280	7216	S	0.0	0.2	0:01.11	systemd-udev
112	root	20	0	2838	1918	1828	S	0.0	0.0	0:00.07	chronyd-starter
113	root	20	0	4472	3116	2860	S	0.0	0.0	0:00.01	cron
114	message+	20	0	8816	5608	4732	S	0.0	0.1	0:00.08	dbus-daemon
121	root	20	0	44996	30856	14712	S	0.0	0.4	0:00.53	networkd-dispat
161	root	20	0	19436	9432	8134	S	0.0	0.1	0:00.16	systemd-logind
188	systemd	20	0	229648	5720	4496	S	0.0	0.1	0:00.13	rsyslogd
244	root	20	0	5492	2764	2564	S	0.0	0.0	0:00.02	getty
288	_chrony	20	0	24888	11872	7224	S	0.0	0.1	0:00.17	chronyd
293	_chrony	20	0	12896	2404	1769	S	0.0	0.0	0:00.00	chronyd
323	root	20	0	123166	32936	16128	S	0.0	0.4	0:00.29	unattended-upgr
378	root	20	0	8648	3512	4668	S	0.0	0.1	0:00.01	login
425	admindh+	20	0	22344	13600	10988	S	0.0	0.2	0:00.23	systemd
438	admindh+	20	0	22912	4128	3116	S	0.0	0.1	0:00.00	(sd-pam)
459	admindh+	20	0	6136	5008	3856	S	0.0	0.1	0:00.06	bash
556	root	20	0	3184	1120	988	S	0.0	0.0	0:00.08	SessionLeader
557	root	20	0	3200	1260	1196	S	0.0	0.0	0:00.03	Relay(561)
561	admindh+	20	0	5764	5536	3864	S	0.0	0.1	0:00.16	bash
895	admindh+	20	0	8164	6856	3904	R	0.0	0.1	0:00.01	top

Result

The program was executed successfully. Linux system and process information was displayed through the required Ubuntu terminal commands, and the process creation demonstration was executed successfully.